



Report

Nara Avila Moraes

Physics Institute
IFUSP

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Summary

① Tasks Status

② RHIC Au–Au-200 + d–Au-200

③ Selected Article

General Tasks Status

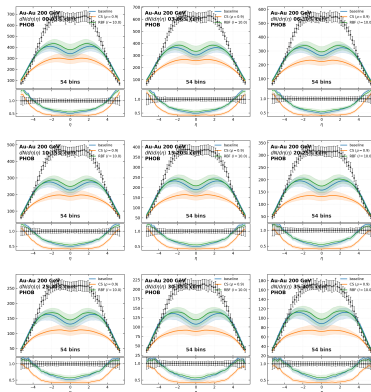
Task	Status	
Correlation Inference Study - Classical Approach		
Correlation Inference Study - Quantum Learning Approach		
2026-11 Paper		Link
2026-12 LASNPA Presentation		Link

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Uncertainty correlation effect	✓	Plots
RHIC Au-Au/d-Au Studies JETSCAPE		
↪ Uncertainty correlation effect	✓	Plots
Atlas ins1759506 studies		
↪ Uncertainty correlation inference	🕒	Notebook
Alice PbPb2760 Scaled Spectra studies		
↪ Uncertainty correlation inference	🕒	Notebook
Synthetic data studies		
↪ Uncertainty correlation inference	🕒	Notebook

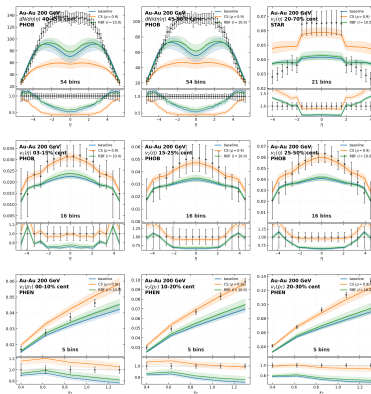
RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors (page 1/4)



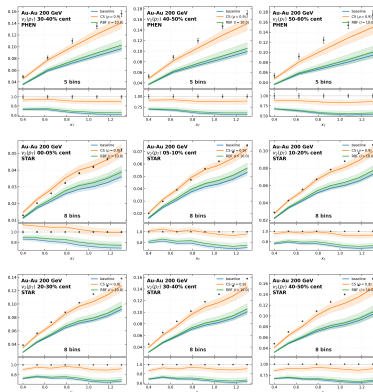
RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors (page 2/4)



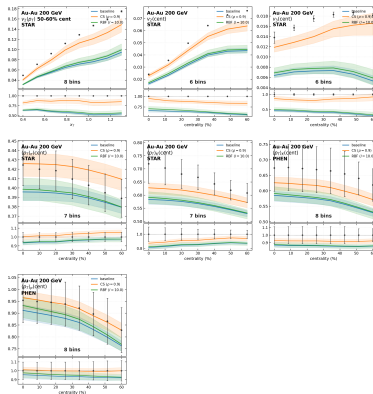
RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors (page 3/4)



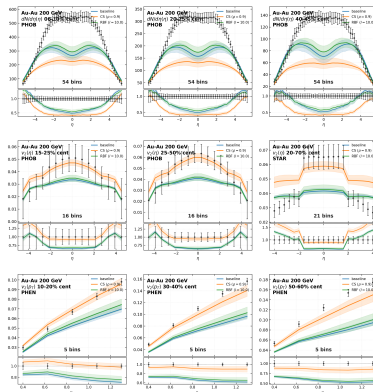
RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors (page 4/4)



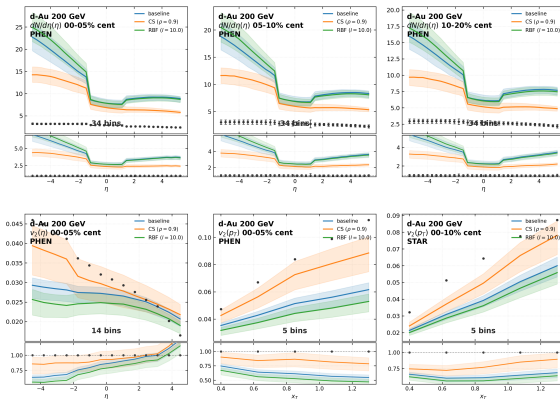
RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors



RHIC Au–Au-200 + d–Au-200

Posterior Predictive Bands from Calibrated Posteriors



Selected Article

Joint Bayesian analysis of soft and high- $p_{\perp}$ probes yields tighter constraints on QGP propertiesMarko Djordjevic,¹ Dusan Zigic,² Igor Salom,² and Magdalena Djordjevic^{2,3,*}¹*Bioinformatics Group, Faculty of Biology, University of Belgrade, Belgrade 11000, Serbia*²*Institute of Physics Belgrade, University of Belgrade, Belgrade 11080, Serbia*³*Serbian Academy of Sciences and Arts, Belgrade 11000, Serbia*

To extract bulk QGP properties, we perform a joint Bayesian calibration of bulk-medium parameters using low- $p_{\perp}$ bulk and high- $p_{\perp}$ tomography within a common medium evolution. Low- $p_{\perp}$ observables are computed with TRENTO+VISHNU; temperature profiles are passed to DREENA-A to predict light/heavy $R_{AA}(p_{\perp})$ and $v_2(p_{\perp})$. Gaussian-process emulation enables Hamiltonian Monte Carlo sampling of the low- $p_{\perp}$ -only and joint posteriors. The low- $p_{\perp}$ -only case underpredicts high- $p_{\perp}$ anisotropy; the joint calibration matches both sectors and markedly tightens bulk-parameter constraints, demonstrating the added power of high- $p_{\perp}$ data.

Figure: Joint bayesian analysis of soft and hard section.